



Kevin Lee 李聖文

Software Development Engineer in Test

Tainan City, Taiwan shengwunlee@gmail.com [linkedin/sheng-wen-lee](https://www.linkedin.com/in/sheng-wen-lee)
[github/shengwun0819](https://github.com/shengwun0819)

ABOUT ME

I'm Kevin, a Software Development Engineer in Test on the CoolWallet team at CoolBitX, with nearly 6 years of software testing experience (since 2020.09, first on the Sygna team and now on the CoolWallet team). I currently focus on API and app functional verification, help the team track and resolve product issues, define testing direction, and independently built an API integration testing project to improve testing efficiency. I also use AI to automatically generate app and API test verification documents, raising both testing efficiency and product delivery quality. Before that, on the Sygna team, I was mainly responsible for product function testing, system integration testing, regression testing configuration, and product performance testing, and I designed and implemented automated testing workflows.

During my bachelor's and master's studies, I focused on projects and research in physiological signals, spectral analysis, and IoT applications. After graduating, I turned that hands-on and research experience into an understanding of system behavior and quality, and have kept sharpening my testing and development skills. Going forward, I want to deepen my expertise in SDET work, test automation, and performance testing, while exploring how AI can be applied to testing workflows.

National University of Tainan — M.S. in Electrical Engineering (Computer Science and Information Engineering Division)

2017 – 2019

Thesis: A Toxic Compound Detection System Combining Mobile Spectral Analysis and Fluorescent Proteins

- Developed spectral analysis techniques in Python, capturing reflectance spectra with a smartphone camera.
- Integrated with the IoTtalk platform to send data to a backend server for toxic compound screening and linear regression analysis.

Kun Shan University — B.S. in Computer Science and Information Engineering

2013 – 2017

Capstone Project: Foot Fitness System

- Built with Arduino and an eHealth module to capture and record foot muscle signals.
- Allring 2016 Innovation & Creativity Competition — Honorable Mention.
- College of Information Science 2016 Project Exhibition — 2nd Place.

CoolBitX · CoolWallet Team — Software Development Engineer in Test Current

2026.05 – Present · Taipei / Tainan, Taiwan

- Responsible for API and app functional verification.
- Help the team track and resolve product issues, and define testing direction.
- Independently developed an API integration testing project to improve testing efficiency.
- Use AI to automatically generate app and API test verification documents, tracking and improving product delivery quality.

CoolBitX · Sygna Team — Software Development Engineer in Test

2020.09 – 2026.05 · Taipei / Tainan, Taiwan

- Performed end-to-end functional verification of products and defined testing direction and strategy.
- Independently developed integration and performance testing projects (PyTest, Jest, K6), cutting overall testing time by about 50%.
- Helped the team track and analyze product issues, identify performance bottlenecks, and verify the effectiveness of fixes.
- Owned verification of third-party service integrations, ensuring Sygna products integrated correctly and reliably with external services.
- Built a Release Regression Testing process using GitHub Actions to automatically trigger PyTest / Jest regression suites, maintaining product quality and release stability.
- Mentored interns on the product's test architecture and collaborated with them on test architecture design.

QA AI Test Case Generation Pipeline

JavaScript · Node.js, Express, Slack API, ClickUp API, Claude CLI (MCP), Ngrok, Tcl/Expect

- Designed and built a fully automated test case generation pipeline: when a ticket moves to a specific status, a Slack webhook detects the event and triggers an AI agent that reads the ticket requirements, generates structured test cases, writes them back to the ClickUp doc library, and replies with the result link in the original thread, replacing manual test case writing by QA.
- Designed a two-stage timeout mechanism (startup timeout / idle timeout) and duplicate-trigger prevention, keeping the service running reliably 24/7 in production.
- Integrated Slack and ClickUp APIs with an Ngrok tunnel to build an observable, maintainable automation service.

Sygnal Hub API System Integration Test

Python, JavaScript, Node.js · pytest, boto3, Ngrok, AWS Cloud, PostgreSQL

- Designed and developed integration testing workflows with Python and pytest, covering core APIs and key scenarios.
- Connected AWS services and test environments to automatically create test data and scenarios, reducing manual setup effort.
- Reduced release testing time by about 50%, stabilizing the release cadence, with 52% test coverage.

Sygnal Bridge / Sygnal Hub System Performance Test

JavaScript, Node.js, Golang (learning) · K6, Ngrok, AWS Cloud

- Designed load and performance test scripts with JavaScript and K6, building a reusable testing framework.
- Planned multiple load scenarios for critical APIs to reproduce client-side issues and traffic patterns.
- Helped the team locate system bottlenecks and verify performance tuning, providing quantitative metrics to guide improvements.

Sygna Bridge API System Integration Test

JavaScript, Node.js · Jest, Ngrok, AWS SDK, DynamoDB, S3

- Designed API test scenarios and maintained test cases and test data together with developers.
- Made the testing process more automated, repeatable, and maintainable, reaching 94.12% test coverage.

Sygna Gate Frontend Test

Python · Selenium · Frontend Regression & E2E Test

- Designed frontend E2E regression test scenarios and test cases based on product features.
- Built a frontend E2E automation project with Python + Selenium (maintained for about 1–2 years).
- Reduced manual effort and regression testing time, strengthening frontend release quality.

SKILLS

Programming & Testing

Python JavaScript Golang (learning)

PyTest K6 Selenium Jest

AI-assisted testing & documentation

Development & Cloud

PostgreSQL MySQL Supabase AWS

GCP Docker Vercel Sentry / Datadog

AI Tools & Automation

Claude Code – AI-assisted coding & test docs

n8n – workflow automation

AI Agent integration (Claude CLI / MCP)

Slack API / ClickUp API

CI/CD & Version Control

Git GitHub GitHub Actions

Scrum Tools

JIRA ClickUp

Manual Testing

Web – Test plans / test cases

API – Design & verification with Postman

APP – Traffic capture & manual verification with Charles Proxy

I'm Kevin, and I hold a master's degree from the Department of Electrical Engineering at National University of Tainan. I currently work as a software test engineer on the CoolWallet team at CoolBitX, responsible for API and app functional verification, tracking and resolving product issues, and defining testing direction. I also use AI to automatically generate app and API test verification documents, continuously improving testing efficiency and product delivery quality. Before that, I spent nearly six years on the Sygna team (2020.09 – 2026.05), responsible for product function testing, system integration testing, regression testing configuration, and product performance testing.

During my bachelor's and master's studies, I focused mainly on physiological signals, spectral analysis, and IoT applications. My undergraduate project, *A Composite Leg Exercise System Controlled by Physiological Information*, dynamically adjusted the exercise resistance model and motion angle based on EMG and body temperature signals. I designed the EMG and temperature signal detectors in C, implemented them on an Arduino UNO WiFi, and recorded users' physiological signals in a cloud database, adjusting exercise parameters in real time to improve training results.

My master's thesis, *A Toxic Compound Detection System Combining Mobile Spectral Analysis and Fluorescent Proteins*, combined Python spectral analysis with a smartphone camera to capture the reflectance spectra of food samples in real time. The data was sent through the IoTtalk platform to a backend server for toxic compound screening and spectral analysis, and linear regression was then used to determine the food's condition, letting users run tests anytime, anywhere.

I also worked as a contract staff member at the Office of Academic Affairs of National Cheng Kung University, handling administrative work and system support. During that time, I built an automation tool for the online document approval workflow with Python and Selenium, turning a manual approval process into an automated one and noticeably improving my colleagues' efficiency.

After joining the Sygna team at CoolBitX, I led or contributed to a number of testing projects:

- Built a Release Regression Testing process with GitHub Actions automatically triggering PyTest / Jest regression suites, ensuring every release has stable, repeatable verification.
- Built an E2E automation project with Selenium, reducing the manual cost of frontend regression testing.

- Developed and maintained API integration tests with PyTest and Jest, cutting overall testing time by about 50%, with test coverage of 94.12% on the Sygna Bridge API and 52% on the Sygna Hub API.
- Set up a K6-based performance testing project to support the team's load testing and performance tuning.
- Worked with third-party service providers on integration verification and quality control.
- Mentored interns on the product's test architecture and collaborated with them on test architecture design.

After transferring to the CoolWallet team in May 2026, my focus shifted to API and app functional verification, while I continue to help the team track and resolve product issues and define testing direction. I developed an API integration testing project to improve overall testing efficiency, and use AI to automatically generate app and API test verification documents to track and improve product delivery quality.

Going forward, I hope to keep working at the intersection of software testing and development: strengthening my SDET skills, expanding my experience in test automation and performance testing, and exploring how AI can be brought into testing workflows, such as test case generation, anomaly detection, and result analysis, to help teams build more efficient and reliable products.